# Logarithmic codimension bounds for midpoint colourings

and certified improvements for the entropic sum–difference problem

B2 continuation from the supplied research handoff

13 October 2026

**Abstract**

For an admissible partition $\mathbb{F}_p^n = A_1 \sqcup \cdots \sqcup A_k \sqcup R$, where $p$ is an odd prime and no pair in a single $A_i$ has midpoint in $R$, a positive-density remainder contains an affine subspace of codimension $O_p(k \log(2k))$. More generally, density at least $\alpha > 0$ gives codimension $O_p(k(\log(2k) + \log(1/\alpha)))$. For $p = 3$ and $\alpha = 1/10$, this improves the $O(k^2)$ bound in the primary source underlying Q4296. The proof combines the established polynomial Freiman–Ruzsa covering theorem with a density-preserving quotient-avoidance lemma. An explicit block construction has largest affine subspace of dimension exactly $n - k$ while its remainder has density at least $9/10$, showing that the upper bound is within a logarithmic factor of the necessary dependence on $k$ in the stated ambient-dimension range. Separately, exact rational logarithm certificates for Q4660 give a finite integer law with entropy-increment ratio strictly larger than 1.40518, improving the supplied handoff's witness near 1.39778. Neither result determines the exact extremal function or the optimal universal entropy constant.

## 1 Scope, baseline, and status

The input is `B2-complete-handoff-2026-10-13.zip`. This continuation concentrates on two targets, without changing the old archive or its stopped run's attempt counts. Q4296 has stable statement identifier `67d9a47fe0b199024fdb`; Q4660 has identifier `4e69214fdfaf0bc4e852`. Exact copies of those catalog records, input hashes, proof receipts, and the source bibliography accompany this note.

For Q4296, the supplied continuation determines the one-colour case exactly: the minimum largest affine-subspace dimension is $n - 1$. It gives a conflict graph formulation but does not establish the general logarithmic bound proved below. The source paper's Theorem 3 proves codimension $O(k^2)$ for the density-1/10 problem over $\mathbb{F}_3$ [1]. For Q4660, the supplied exact three-point witness has ratio between 1.397784606785787064840970275371855 51 and 1.397784606785787064840970275371855 52 [4].

**Status.** The combinatorial conclusions below have complete written proofs using the established bounded-torsion polynomial Freiman–Ruzsa theorem [2]. The entropy conclusions have exact rational certificates. Finite controls check elementary proof components, not all dimensions and not the external PFR theorem. There has been no external peer review or formal proof-assistant verification. These are advances relative to the supplied handoff and the checked primary statements; worldwide priority has not been established. The exact targets Q4296 and Q4660 remain open within this continuation.

## 2 An explicit covering-to-subspace transfer theorem

All logarithms in the combinatorial proof are natural unless their base is shown. An affine subspace is nonempty. Empty colour classes are allowed in intermediate restrictions; this only strengthens applicability to the original partition problem.

Fix an odd prime $p$. Let $M_p(k)$ be any positive integer such that every nonempty set $A$ in a finite-dimensional $\mathbb{F}_p$-vector space with $|A+A| \le k|A|$ is contained in at most $M_p(k)$ cosets of a subspace $H$ with $|H| \le |A|$. The established PFR theorem provides an absolute constant $C_0$ for which one may take

$$M_p(k) = \left\lceil (2k)^{C_0 p^3} \right\rceil. \tag{1}$$

This is the covering form of Theorem 1.1 in [2], not a conjectural polynomial Bogolyubov statement.

**Theorem 2.1** (Quantitative transfer)**.** *Let $k \ge 1$, $0 < \alpha \le 1$, and suppose*

$$\mathbb{F}_p^n = A_1 \sqcup \cdots \sqcup A_k \sqcup R, \qquad |R| \ge \alpha p^n,$$

*with $(x+y)/2 \notin R$ for every $x, y \in A_i$ and every $i$. Put*

$$M = M_p(k), \qquad s = \left\lceil \log_p(2kM) \right\rceil, \tag{2}$$

$$\tau = \frac{1}{4p^s}, \qquad T = \left\lceil (2k+1) \log \frac{4p^s}{\alpha} \right\rceil. \tag{3}$$

*Then $R$ contains an affine subspace of codimension at most*

$$\min\{n, T + ks\}.$$

*In particular, for each fixed odd prime $p$ the codimension is*

$$O_p\left(k\left[\log(2k) + \log(1/\alpha)\right]\right). \tag{4}$$

*Using the $p$-dependence in* (1)*, an absolute implied constant gives the bound $O(k[p^3 \log(2k) + \log(1/\alpha)])$.*

**Corollary 2.2** (Q4296)**.** *There is an absolute constant $C$ such that every admissible partition in Q4296 has an affine subspace $V \subseteq R$ satisfying*

$$\dim V \ge \max\{0, n - Ck\log(2k)\}.$$

*Thus the established quadratic codimension loss can be replaced by a logarithmic overhead over linear dependence on the number of colours.*

The essential distinction is between covering a colour by polynomially many cosets and enclosing it in one affine span. The former can be avoided at low codimension while preserving the density of the other colours. The following lemma supplies that preservation.

## 2.1 Density-preserving avoidance in a quotient

**Lemma 2.3** (Weighted avoidance)**.** *Let $W$ be an $\mathbb{F}_p$-vector space, $H \le W$, and suppose that $A \subseteq W$ is contained in at most $M$ cosets of $H$. Let $1 \le s \le \dim(W/H)$ and $Mp^{-s} < 1$. For every function $f : W \longrightarrow [0, \infty)$ there is an affine subspace $W' \subseteq W$ of codimension $s$ such that*

$$W' \cap A = \varnothing, \qquad \mathbb{E}_{x \in W'} f(x) \le \frac{\mathbb{E}_{x \in W} f(x)}{1 - Mp^{-s}}. \tag{5}$$

*The same assertion holds after translating the ambient space.*

*Proof.* Write $Q = W/H$, let $\pi : W \to Q$ be the quotient map, and choose a set $S \subseteq Q$ with $|S| \le M$ such that $A \subseteq \pi^{-1}(S)$. Choose uniformly a codimension-$s$ affine subspace $U$ of $Q$. Every point of $Q$ belongs to $U$ with probability $p^{-s}$. Therefore the event $\mathcal{G} = \{U \cap S = \varnothing\}$ satisfies

$$\mathbb{P}(\mathcal{G}) \ge 1 - |S|p^{-s} \ge 1 - Mp^{-s} > 0.$$

Set $Z(U) = \mathbb{E}_{x \in \pi^{-1}(U)} f(x)$. All pullbacks have the same size, $|W|p^{-s}$, so uniform point inclusion gives

$$\mathbb{E}Z(U) = \mathbb{E}_{x \in W} f(x).$$

Since $Z \ge 0$,

$$\mathbb{E}[Z \mid \mathcal{G}] = \frac{\mathbb{E}[Z\mathbf{1}_{\mathcal{G}}]}{\mathbb{P}(\mathcal{G})} \le \frac{\mathbb{E}_{x \in W} f(x)}{1 - Mp^{-s}}.$$

Some successful $U$ has at most this conditional mean. Its pullback is the required $W'$. In particular, the proof needs no independence between avoidance and the density of $f$. □

Taking $f$ to be the indicator of the union of all surviving colours, and choosing $Mp^{-s} \le 1/(2k)$, incurs only the density factor $(1 - 1/(2k))^{-1}$. This factor, rather than a fixed factor larger than one, is what can be iterated $k$ times without an exponentially small initial coloured density.

## 2.2 The density increment and its odds potential

**Lemma 2.4** (Hyperplane increment)**.** *Suppose $A, R \subseteq \mathbb{F}_p^m$ have densities $\beta > 0$ and $a > 0$, and $(A + A) \cap (2R) = \varnothing$. Then some affine hyperplane has $R$-density at least $a(1 + \beta/2)$.*

*Proof.* Use normalized Fourier coefficients. Orthogonality and the disjointness hypothesis imply

$$0 = a\beta^2 + \sum_{\chi \ne 1} \widehat{\mathbf{1}_A}(\chi)^2 \overline{\widehat{\mathbf{1}_{2R}}(\chi)}.$$

Parseval gives $\sum_{\chi \ne 1} |\widehat{\mathbf{1}_A}(\chi)|^2 = \beta(1 - \beta) \le \beta$. Consequently some nontrivial character has $|\widehat{\mathbf{1}_{2R}}(\chi)| \ge a\beta$. Let $a_0, \dots, a_{p-1}$ be the densities of $2R$ in the cosets of $\ker \chi$. Their mean is $a$. With $u_j = a_j - a$, one has $\sum_j u_j = 0$, hence

$$|\widehat{\mathbf{1}_{2R}}(\chi)| \le \frac{1}{p} \sum_j |u_j| = \frac{2}{p} \sum_{u_j > 0} u_j \le 2 \max_j u_j.$$

Thus some coset has density at least $a + a\beta/2$. Applying multiplication by $2^{-1}$ gives an affine hyperplane with the same density for $R$. □

The $p = 3$ increment is the elementary Fourier input of the primary source [1]; the formulation above covers every odd prime. The following potential makes the dependence on the starting density explicit.

**Lemma 2.5** (Odds growth)**.** *In an admissible partition with at most $k$ colours and remainder density $a \in (0, 1)$, there is a hyperplane restriction whose new remainder density $a'$ either equals one or satisfies*

$$\frac{a'}{1 - a'} \ge \left(1 + \frac{1}{2k}\right) \frac{a}{1 - a}. \tag{6}$$

*Proof.* The largest colour has density at least $(1 - a)/k$. By Lemma 2.4, $a' \ge a(1 + (1 - a)/(2k))$. Since $x/(1 - x)$ is increasing on $(0, 1)$,

$$\frac{a'/(1 - a')}{a/(1 - a)} \ge \frac{1 + (1 - a)/(2k)}{1 - a/(2k)} = 1 + \frac{1}{2k - a} \ge 1 + \frac{1}{2k}.$$

A full hyperplane terminates the process instead. □

### 2.3 Removing all colours at polynomially small density

**Lemma 2.6** (High-density clearance)**.** *With $M, s, \tau$ from (2)–(3), let an admissible partition of an affine space $W$ have total coloured density at most $\tau$. Then $R \cap W$ contains an affine subspace of relative codimension at most $ks$.*

*Proof.* Translate $W$ to a vector space as needed. Suppose the current ambient space is $W_j$, its coloured union is $B_j$, and its coloured density is $\delta_j$. If $B_j$ is empty, stop. Otherwise choose a largest nonempty colour $A$ among the surviving labels. Admissibility gives

$$|A + A| \le |B_j| \le k|A|.$$

The first inequality holds because doubling is a bijection on an odd-characteristic vector space and $A + A$ avoids twice the remainder. PFR covers $A$ by at most $M$ cosets of a subspace $H$ satisfying

$$|H| \le |A| \le \delta_j |W_j|.$$

Apply Lemma 2.3 with $f = \mathbf{1}_{B_j}$ and the fixed codimension $s$. It removes $A$ completely and yields

$$\delta_{j+1} \le \frac{\delta_j}{1 - Mp^{-s}} \le \frac{\delta_j}{1 - 1/(2k)}. \tag{7}$$

For completeness, the required quotient dimension is available at every step; it is not an additional assumption. Inductively, after at most $k$ steps,

$$\delta_j \le \tau \left(1 - \frac{1}{2k}\right)^{-j} \le 2\tau = \frac{1}{2p^s}.$$

The second inequality follows from Bernoulli's inequality $(1 - 1/(2k))^k \ge 1/2$. Therefore $|H|/|W_j| \le 1/(2p^s) < p^{-s}$, so $\dim(W_j/H) > s$. In particular, any space too small for this restriction must already have no coloured points. This also makes the induction in (7) noncircular: the bound at step $j$ validates the next application.

Each successful restriction removes at least one colour label. After at most $k$ restrictions no labels remain, and the total codimension cost is at most $ks$. □

### 2.4 Completion of the transfer proof

*Proof of Theorem 2.1.* If $R$ is the entire ambient space, there is nothing to prove. Otherwise iterate Lemma 2.5 until the remainder has density at least $1 - \tau$. Admissibility is inherited by affine restriction because a midpoint of points in an affine space remains in that space. The initial odds are at least $\alpha$. Also

$$\log\left(1 + \frac{1}{2k}\right) \ge \frac{1}{2k+1}.$$

If no earlier restriction is full, after $T$ steps the odds are at least

$$\alpha \exp\left(\frac{T}{2k+1}\right) \ge 4p^s = \tau^{-1}.$$

The coloured density is then at most $1/(1 + \tau^{-1}) < \tau$. If dimension reaches zero earlier, positive remainder density means the single point is already in $R$.

Apply Lemma 2.6 in the resulting affine space. The total cost is at most $T + ks$. This proves the explicit bound. Under (1), $\log M = O(p^3 \log(2k))$, and substitution into (2)–(3) yields the stated asymptotic bounds. □

*Remark* 2.7 (What changed in the proof mechanism). A constant density inflation at each of $k$ eliminations would require an exponentially small coloured density at the start of clearance. The weighted avoidance lemma instead sets each failure probability to at most $1/(2k)$, so the total inflation is at most two. PFR then requires only a polynomially small coloured density. Reaching that threshold takes $O_p(k \log k)$ hyperplane increments, and clearance also costs $O_p(k \log k)$. Merely replacing an old theorem by PFR without controlling these density losses would not establish the improvement.

# 3 A deterministic construction with exact linear loss

**Proposition 3.1** (Block-zero construction)**.** *Let $p$ be an odd prime, $k, b \geq 1$, and $n \geq kb$. There is an admissible partition of $\mathbb{F}_p^n$ into $k$ colours and a remainder $R$ for which*

$$\frac{|R|}{p^n} = (1 - p^{-b})^k, \qquad \max\{\dim V : V \subseteq R \text{ affine}\} = n - k.$$

*In particular, taking $b = \lceil \log_p(10k) \rceil$ gives density at least $9/10$ and exact loss $k$ whenever $n \geq k\lceil \log_p(10k) \rceil$.*

*Proof.* Write a point as $(x^{(1)}, \ldots, x^{(k)}, y)$, with $x^{(i)} \in \mathbb{F}_p^b$ and $y \in \mathbb{F}_p^{n-kb}$. Put

$$R = \{x : x^{(i)} \neq 0 \text{ for every } i\},$$

and colour every point outside $R$ by the index of its first zero block. If two points have colour $i$, their $i$th blocks are zero, as is that of their midpoint. The midpoint is therefore outside $R$, irrespective of its other blocks. This proves admissibility. Counting independent blocks gives the stated density.

Let $V \subseteq R$ be affine. Its image under each block projection is an affine subspace $a_i + L_i \subseteq \mathbb{F}_p^b$ avoiding zero. Thus $a_i \notin L_i$, and some linear functional $\ell_i$ vanishes on $L_i$ and satisfies $\ell_i(a_i) = 1$. The pullbacks of these $k$ functionals to the ambient space are linearly independent, because they are nonzero and have disjoint coordinate supports. Consequently $V$ satisfies $k$ independent affine linear equations and $\dim V \leq n - k$.

Conversely, requiring the first coordinate in each block to equal one leaves an affine subspace of dimension $n-k$ wholly inside $R$. Finally, Bernoulli's inequality gives $(1-p^{-b})^k \geq 1-kp^{-b} \geq 9/10$ for the chosen $b$. □

This is an explicit companion to the lower-construction direction already noted in the primary source, not a claim to have discovered the necessity of linear loss for the first time. Together with Corollary 2.2, it leaves a factor $O(\log k)$ between necessary and sufficient codimension dependence, for the ambient dimensions covered by the construction. It does not determine the exact sharp function of $(n, k)$.

# 4 Certified entropy-ratio improvement for Q4660

Let $X'$ be an independent copy of a discrete real-valued random variable $X$ with finite Shannon entropy. Write

$$\Delta_+(X) = H(X + X') - H(X), \qquad \Delta_-(X) = H(X - X') - H(X).$$

Entropy may be measured in nats or bits; the ratio $\mathcal{R}(X) = \Delta_-(X)/\Delta_+(X)$ is unchanged. The known universal inequality is $\Delta_-(X) \leq 2\Delta_+(X)$, while sharpness of two is the question recorded in Q4660 [3]. Let $C_*$ denote the least universal constant.

**Proposition 4.1** (Two exact finite witnesses)**.** *For the law on* $\{0,1,2,3,4\}$ *with probability vector*

$$\frac{1}{10000}(8830, 1009, 143, 16, 2),$$

*one has*

$$\begin{aligned} 1.405025826013192214095918189356 &< \mathcal{R}(X) \\ &< 1.405025826013192214095918189357. \end{aligned}$$

*For the law on* $\{0,1,\dots,7\}$ *with denominator* $10^8$ *and numerators*

$$\begin{aligned} (88162304,\ 10154519,\ 1485728,\ 177393,\\ 18242,\ 1665,\ 138,\ 11), \end{aligned}$$

*one has*

$$\begin{aligned} 1.405180966075633572545128347101 &< \mathcal{R}(X) \\ &< 1.405180966075633572545128347102. \end{aligned}$$

*In particular,*

$$\boxed{C_* > \frac{70259}{50000} = 1.40518.} \tag{8}$$

*The supplied handoff's lower witness is strictly improved.*

*Exact certificate and its mathematical justification.* For integer probability weights $w_i$ with total $D$, the output weights are obtained without rounding:

$$s_t = \sum_{i+j=t} w_i w_j, \qquad d_t = \sum_{i-j=t} w_i w_j.$$

Each output has total weight $D^2$. Entropy of any count vector $c$ with total $N$ is exactly

$$H(c) = \log N - \frac{1}{N} \sum_{c_i>0} c_i \log c_i. \tag{9}$$

All logarithms in the certificate are enclosed by rational bounds. For $1 \le r \le 2$, put $t = (r-1)/(r+1) \in [0, 1/3]$. Then

$$L_J(r) = 2 \sum_{j=0}^{J-1} \frac{t^{2j+1}}{2j+1}, \tag{10}$$

$$0 \le \log r - L_J(r) \le \frac{2t^{2J+1}}{(2J+1)(1-t^2)}. \tag{11}$$

The upper bound follows by bounding every denominator in the positive tail below by $2J+1$ and summing the resulting geometric series. For a positive integer $m$, write $m = 2^e r$ with $1 \le r < 2$ and apply these bounds to $\log m = e \log 2 + \log r$.

The supplied script `certify_entropy.py` uses $J = 90$, computes the series and tail with exact fractions, and rounds each logarithm interval outward to a grid of mesh $10^{-65}$. Substitution in (9) and outward interval subtraction give positive lower bounds for both increments. If $L_\pm \le \Delta_\pm \le U_\pm$ with $L_+ > 0$, then

$$\frac{L_-}{U_+} \le \mathcal{R}(X) \le \frac{U_-}{L_+}.$$

The two resulting enclosing intervals lie strictly between the displayed endpoints. The script also verifies the positive rational lower bound

$$L_- - \frac{70259}{50000}U_+ > 0$$

for the eight-point law, and the analogous inequality with 281/200 for the five-point law. All weights, interval endpoints, and positive-gap receipts are retained in `ENTROPY-CERTIFICATE.json`. The displayed values are therefore exact enclosures, not optimizer output. □

### 4.1 The witness survives arbitrarily large input entropy

**Corollary 4.2.** *For every $h > 0$ there is a positive-integer-valued $Y$ with $H(Y) > h$ and $\mathcal{R}(Y) > 1.40518$.*

*Proof.* Let $Z$ have the eight-point law, let $Z_0, \dots, Z_{m-1}$ be independent copies, and put

$$Y = 1 + \sum_{j=0}^{m-1} 16^j Z_j.$$

The digit encodings for the input and the sums are injective: input digits lie in $[0, 7]$ and sum digits in $[0, 14]$. Difference digits lie in $[-7, 7]$ and also have a unique base-16 encoding. Indeed, a difference between two such encodings has digits in $[-14, 14]$; a nonzero highest term cannot be cancelled by all the lower terms because $14 \sum_{j<r} 16^j < 16^r$.

Thus all relevant vector entropies are preserved by encoding. Coordinate independence gives

$$H(Y) = mH(Z), \qquad \Delta_+(Y) = m\Delta_+(Z), \qquad \Delta_-(Y) = m\Delta_-(Z).$$

The ratio is unchanged, while $mH(Z)$ is arbitrarily large. □

This use of tensorization does not amplify a fixed ratio toward two. It shows that the improved obstruction persists at every entropy scale. It does not improve the universal upper constant or the separate sum–product target Q4659.

## 5 Verification, limitations, and the next mathematical obstacle

### 5.1 Reproduction

Run the following from the extracted result directory with Python 3.10 or later. The certificate scripts use only the standard library.

```
python certify_entropy.py
python check_midpoint_lemmas.py
python check_block_construction.py
```

The midpoint checker examines all 282339 labelled partitions of $\mathbb{F}_3^2$ using one, two, or three colour labels plus the remainder. Among them it checks the increment and odds inequalities for all 29706 admissible partitions with nonempty proper remainder. The quotient tests check all 285184 allowed cover/test-set pairs in $\mathbb{F}_3^2$, together with 6684 nonnegative-weight controls. A further 9900 rational odds checks and 100 cumulative-loss checks pass. These are exact integer or fraction comparisons.

The block-construction checker enumerates every affine subspace in each of eight finite cases. In dimensions three and four, the lists contain 184 and 2452 affine subspaces respectively. Every case attains exactly $n - k$ as predicted. None of these finite checks is offered as a substitute for the all-size block proof or Theorem 2.1.

The entropy certificate itself is a finite exact proof for the displayed laws. The optional numerical exploration script only explains discovery; it is not needed for verification, and its local optimizations are not claims of global optimality.

### 5.2 Audit of the critical proof obligations

The general midpoint proof has four particularly important points. The small-doubling estimate uses the *total* coloured union in the current affine space, not the size of the selected colour alone. The quotient dimension is justified from $|H| \leq |A|$ and the running density invariant before every avoidance step. Conditional averaging uses nonnegativity and uniform point inclusion, not an unjustified independence assumption. Finally, removal of a label is permanent under later nested restrictions, so there are at most $k$ clearance steps.

The argument is an existence proof. It does not supply a new polynomial-time algorithm for finding the PFR covers, a numerical value of the absolute PFR exponent, or a sharp finite-$n$ codimension constant. Small ambient dimensions are covered by the nonempty-point stopping case.

### 5.3 Remaining gaps

The exact value of the Q4296 extremal function is not determined. The block construction supplies necessary linear loss, while the new theorem supplies sufficient $O(k \log k)$ loss. The remaining structural target is a clearance/density-increment mechanism with an amortized constant codimension cost per colour, or a construction proving that a superlinear loss is actually necessary. Improving constants inside the present PFR input alone does not remove the logarithm, because the quotient avoidance step still costs logarithmically in the number of covering cosets.

For Q4660, the certified lower bound is not a universal upper bound and is not a proof that two is sharp. The local search over short arithmetic progressions leaves arbitrary supports and arbitrary probability laws untreated. The correct catalog disposition of both targets is therefore *partial progress*, with the exact proofs and scopes above retained.

## References

[1] Tomasz Kościuszko, *Arithmetic Progressions in Midpoint Colourings*, arXiv:2610.11847v1, submitted 8 October 2026. Problem 4, Lemma 1, and Theorem 3. `https://arxiv.org/html/2610.11847v1`.

[2] W. T. Gowers, Ben Green, Freddie Manners, and Terence Tao, *Marton's conjecture in abelian groups with bounded torsion*, arXiv:2404.02244v2, 20 May 2024, Theorem 1.1. `https://arxiv.org/html/2404.02244v2`.

[3] Rupert Li, *The Entropic Sum-Product Phenomenon*, arXiv:2607.29042v1, submitted 31 July 2026, Section 8. `https://arxiv.org/html/2607.29042v1`.

[4] User-supplied *B2 complete research handoff*, 13 October 2026. The catalog snapshot and relevant attempt and certificate files are identified by exact paths and hashes in the accompanying `PROVENANCE.json`.